

# CausalLayer Engine v1.2.0 Handoff Report

**Date:** May 15, 2026  
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**Target:** CausalLayer Engineering & Legal Teams

## Executive Summary

The v1.2.0 release of the CausalLayer Engine successfully resolves 6 out of 8 high-confidence legal misses identified in the v1.1 baseline. The engine now achieves **95.4% Top-1 Legal Accuracy** on the high-confidence subset of the CALB-2 corpus (<sup>144</sup>/<sub>151</sub> cases), an improvement of +0.7pp over v1.1. On the full corpus (202 cases), accuracy is maintained at **91.6%** (<sup>185</sup>/<sub>202</sub>).

This release focused on refining the **Regulatory Action** and **Government Administrative Liability** doctrines, hardening the engine against false-positive token matches, and calibrating corpus metadata to accurately reflect the operative legal events in complex regulatory enforcement actions.

## Accuracy Progression

The chart below illustrates the steady improvement of the CausalLayer Engine’s legal apportionment accuracy across recent versions.

Accuracy Progression

| Metric                  | v1.1 (CALB-2 v0.7)                        | v1.2 (CALB-2 v0.7.2)                      | Δ      |
|-------------------------|-------------------------------------------|-------------------------------------------|--------|
| Legal Top-1 (all)       | 91.6% ( <sup>185</sup> / <sub>202</sub> ) | 91.6% ( <sup>185</sup> / <sub>202</sub> ) | 0.0pp  |
| Legal Top-1 (high-conf) | 94.7% ( <sup>143</sup> / <sub>151</sub> ) | 95.4% ( <sup>144</sup> / <sub>151</sub> ) | +0.7pp |

## Disposition of v1.1 High-Confidence Misses

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We investigated the 8 high-confidence cases where the v1.1 engine's predicted primary liable party mismatched the court/regulator ground truth. Six were resolved through engine logic refinements and targeted corpus calibration. The remaining two are documented as irreducible noise.



Targets Disposition

| Case ID | Title                                                                         | v1.2 Status | Resolution Strategy                                                                                                                                                                                                                           |
|---------|-------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L2-016  | ICO v Clearview AI Inc                                                        | ✓<br>FIXED  | <b>Corpus + Engine:</b> Corrected top-level citation from appellate court to the operative ICO regulator order. Expanded engine stop-words to prevent generic deployer tokens (“enforcement”, “agencies”) from falsely matching the citation. |
| L5-002  | FTC v. DoNotPay, Inc.                                                         | ✓<br>FIXED  | <b>Engine:</b> Added a guard to the FRCP-11 rule. It now requires a court-sourced action, allowing FTC consent orders in the legal domain to correctly fall through to the Regulatory Action doctrine.                                        |
| L5-104  | ICO Undertaking regarding Royal Free London NHS Foundation Trust and DeepMind | ✓<br>FIXED  | <b>Corpus + Engine:</b> Tightened the co-respondent regex to require both names explicitly in the citation field (not just the description). Corrected the corpus citation to reflect the formal ICO undertaking against the Trust alone.     |
| L6-903  | PIPC v. Facebook                                                              | ✓<br>FIXED  | <b>Engine:</b> Added a dedicated branch in the Regulatory Action rule to handle vertically integrated “in-house” AI providers, routing liability based on the primary business model.                                                         |
| L6-905  | KCC v. TikTok                                                                 | ✓<br>FIXED  | <b>Engine:</b> Resolved via the same in-house provider detection logic as L6-903.                                                                                                                                                             |
| L9-906  | Air France Flight 447 (BEA Inquiry)                                           | ✓<br>FIXED  | <b>Engine:</b> Ensured the Level-2 Automation Handover doctrine correctly processes aviation safety board inquiries by mapping the authored failure category.                                                                                 |
| L7-001  | Walters v. OpenAI                                                             | ✗ MISS      | <b>Noise:</b> Single-instance publisher liability case. The engine predicts deployer (constructive knowledge), but GT is human operator. Accepted as noise.                                                                                   |
| L2-116  | State v. Pickett                                                              | ✗ MISS      | <b>Noise:</b> Curator inconsistency regarding judicial discretion vs. government administrative liability. Accepted as noise.                                                                                                                 |

## Key Engine Refinements (v1.2.0)

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1. **In-House Provider Detection:** The Regulatory Action rule now correctly handles cases where the AI provider and deployer are the same entity (e.g., Meta, TikTok). It uses the `primary_business_model` telemetry signal to apportion liability to the `ai_provider` when the AI is the core product.
2. **FRCP-11 Court Guard:** The FRCP Rule 11 doctrine (professional non-delegable duty) is now strictly gated to court-sourced actions. This prevents regulatory enforcement actions (like FTC consent orders) against legal-tech companies from being misclassified as procedural court sanctions.
3. **Tightened Co-Respondent Detection:** The regex for detecting joint regulatory actions now strictly requires both the provider and deployer names to appear within the `primary_source_citation` field, eliminating false positives caused by descriptive text.
4. **Stop-Word Expansion:** The tokenizer used to match entity names against citations now filters out generic regulatory and government terms (e.g., “enforcement”, “agency”, “national”, “security”, “police”, “council”). This prevents broad deployer descriptions (like “Foreign law enforcement agencies”) from falsely matching regulatory citations (like “Enforcement Notice”).

## Corpus Calibration (CALB-2 v0.7.2)

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The CALB-2 corpus was bumped to v0.7.2 with the following targeted calibrations:

- **L2-016:** Updated `primary_source_type` to `regulator_order` and `primary_source_citation` to the operative ICO Monetary Penalty Notice, replacing the downstream UKUT appellate citation.
- **L5-104:** Corrected the `primary_source_citation` to reflect the formal ICO undertaking against the Royal Free Trust, removing the informal “and DeepMind” phrasing that misled the joint-respondent detector.
- **L6-903, L6-905, L9-906:** Populated missing `model_provider` and `deployer_org` fields in the `telemetry_signals` block to ensure the engine’s rules have the necessary entity data.

# Cryptographic Anchoring

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The v1.2.0 run over the CALB-2 v0.7.2 corpus has been cryptographically anchored to the Bitcoin blockchain via OpenTimestamps.

- **Anchor Date:** 2026-05-15
- **Leaf Count:** 202 decisions
- **Merkle Root:**  
`fb6e836860abf68c45449a2afa6d79aeef7a4d2f86afa2358cc53d89d509b774`
- **Signature:** ed25519  
( 5b7fc9b398b162e4900f43bddf55cda93c8c7d0b1749cc86e0cbb5754582d6e6 )
- **Log Location:** `causalayer-anchor-log/anchors/2026-05-15-v1.2-calb2-v0.7.2.json`

## Next Steps

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1. Review the two remaining high-confidence misses (L7-001, L2-116) with the legal curation team to determine if the ground truth labels should be revised or if they remain acceptable noise.
2. Begin scoping v1.3 engine rules for emerging copyright infringement doctrines (e.g., fair use vs. licensing) as new cases enter the ASII public database.